CHAPTER 5

# Complex numbers and geometry

Angles and ratios of lengths, $n$-th roots of unity

**Objectives.** Interpret the modulus and an argument of $\frac{c-a}{b-a}$; prove that points are collinear, that lines are perpendicular, the nature of a triangle; determine sets of points; determine the $n$-th roots of unity and use them for regular polygons.

**Prerequisites.** Chapters 2, 3 and 4; oriented angles of vectors; special triangles.

## Fluency

**Fluency · No calculator, five minutes.**

1. Give an argument of $-1+i$ and of $-3i$.
2. Write $\mathrm{e}^{2i\pi/3}$ in algebraic form.
3. If $A$ and $B$ have affixes $a$ and $b$, what is the affix of $\overrightarrow{AB}$?
4. Compute $\dfrac{1+i}{1-i}$.
5. At most how many solutions does the equation $z^5 = 1$ have in $\mathbb{C}$?
6. What is the geometric effect of multiplication by $\mathrm{e}^{i\pi/3}$?